ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE H

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Proximity and optical sensors and system block diagrams, PLC inputs, outputs and simple ladder rungs, binary numbers, two-cylinder pneumatic sequences and time-delay valves, motor types and reversing a DC motor, and workshop housekeeping and replacing a solenoid valve.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all symbols, diagrams and circuits neatly, using a ruler.
5. Use only a black or blue pen. Write neatly and legibly.

QUESTION 1: SENSORS AND SYSTEMS

1.1 Explain the difference between an inductive and a capacitive proximity sensor, and give ONE material that each can detect. (4)

1.2 Name and briefly describe TWO types of optical (photoelectric) sensors. (4)

1.3 Draw a block diagram (input, process, output) of an automatic washing machine, and give ONE example of each block. (6)

1.4 Explain why sensors are needed in a mechatronic system. (2)

1.5 Give TWO examples of transducers and the energy change each makes. (4)

[20]

QUESTION 2: PLC BASICS

2.1 Write PLC in full and explain what a PLC is. (3)

2.2 Give TWO examples of PLC input devices and TWO examples of PLC output devices. (4)

2.3 Draw a ladder rung in which output Q0.0 switches on only when input I0.0 AND input I0.1 are both on. (3)

2.4 Draw a ladder rung in which output Q0.1 switches on when input I0.2 OR input I0.3 is on. (3)

2.5 Give TWO advantages of a PLC over hard-wired relay control. (4)

2.6 Convert the decimal number 25 to binary. (3)

[20]

QUESTION 3: PNEUMATIC SEQUENCES

3.1 Two double-acting cylinders, A and B, must work in the sequence A+ B+ A− B−. Draw the displacement-step diagram for this sequence. (8)

3.2 Explain what the "+" and "−" signs mean in a sequence such as A+ B+ A− B−. (2)

3.3 Explain what a pneumatic time-delay valve does and give ONE use. (4)

3.4 Name THREE ways in which a pneumatic valve can be operated (actuated). (3)

3.5 Explain why roller-lever valves (limit valves) are used in an automatic pneumatic sequence. (3)

[20]

QUESTION 4: MOTORS

4.1 Name FOUR types of electric motors used in mechatronic systems. (4)

4.2 Explain how the direction of rotation of a permanent-magnet DC motor is reversed. (2)

4.3 Draw a circuit in which two relays (K1 forward and K2 reverse) reverse the polarity of the supply to a DC motor. (6)

4.4 Explain the difference between a stepper motor and a servo motor. (4)

4.5 Give FOUR items of information found on the nameplate (rating plate) of a motor. (4)

[20]

QUESTION 5: WORKSHOP PRACTICE AND MAINTENANCE

5.1 Give FOUR reasons why good housekeeping is important in a mechatronics workshop. (4)

5.2 Name FOUR hand tools used to maintain mechatronic equipment, and give the use of each. (4)

5.3 Describe, step by step, how to replace a faulty solenoid valve on a pneumatic machine safely. (6)

5.4 Explain the difference between a planned and an unplanned stoppage of a machine. (4)

5.5 What is the purpose of a danger tag? (2)

[20]

TOTAL: 100

END OF QUESTION PAPER